

Logos Experiment 17

Large Scaling + 10k Training | 124M Decoder-Only Language Model

1. Experiment Purpose

Experiment 17 extends the large-scale Logos model from 7,500 to 10,000 training iterations while keeping the model architecture, dataset, batch size, context length, tokenizer, and training objective unchanged. The goal is to test whether the 124M-parameter Logos model continues improving or begins to plateau under longer training.

This is a controlled training-duration experiment. No architectural or dataset change was introduced compared with Experiment 16.

2. Configuration

Component	Value
Run name	logos_exp17_large_scaling_weight_tying_swiglu_10k_train
Dataset	OpenWebText subset, 100k samples
Tokenizer	GPT-2 BPE tokenizer
Architecture	Decoder-only GPT-style Transformer
Layers	12
Embedding dimension	768
Attention heads	12
Head dimension	64
Context length	512 tokens
Batch size	32
Tokens per step	16,384
FFN type	SwiGLU
Weight tying	True
Training iterations	10,000
Parameters	124,031,232

3. Final Results

Metric	Experiment 17 Result
Best iteration	10,000
Best validation loss	4.1182
Final validation loss	4.1167
Final validation perplexity	61.36
Checkpoint size	473.20 MB
Peak VRAM	17.73 GB
Training time	2263.57 sec = 37.73 min
Generation speed	67.68 tokens/sec
Generation latency	14.77 ms/token
Average GPU power	292.37 W
Estimated GPU energy	183.67 Wh

4. Comparison Against Experiment 16

Metric	Exp 16: 7500 iters	Exp 17: 10k iters	Change
Parameters	124.03M	124.03M	Same
Dataset	100k	100k	Same
Context	512	512	Same
Best val loss	4.2301	4.1182	Improved
Final val loss	4.2254	4.1167	Improved
Final perplexity	68.40	61.36	Improved
Checkpoint	473.20 MB	473.20 MB	Same
Peak VRAM	17.73 GB	17.73 GB	Same
Training time	28.94 min	37.73 min	Higher
Energy	141.11 Wh	183.67 Wh	Higher

5. Scientific Interpretation

Experiment 17 is the strongest Logos result so far. The model achieved a final validation perplexity of 61.36, improving from 68.40 in Experiment 16. Since the best iteration was still the final training checkpoint, the model may not be fully plateaued yet.

The improvement was achieved without increasing parameter count, checkpoint size, VRAM usage, architecture complexity, or context length. The only major cost was additional training time and energy consumption, which is expected from increasing the training budget from 7,500 to 10,000 iterations.

6. Scaling Trend

Run	Params	Dataset	Iterations	Final Val Loss	PPL
Exp 11: compact WT + SwiGLU	19.24M	10k	3000	5.3123	202.82
Exp 13: medium WT + SwiGLU	51.20M	50k	3000	5.0421	154.79
Exp 15: large WT + SwiGLU	124.03M	100k	5000	4.4663	87.03
Exp 16: large extended	124.03M	100k	7500	4.2254	68.40
Exp 17: large 10k	124.03M	100k	10000	4.1167	61.36

The trend shows clear improvement from compact to medium and large models, and continued improvement from longer training at fixed parameter count. This supports the project's main claim that capability improves when efficient architecture choices are combined with scaling and extended training.

7. Conclusion

Experiment 17 is currently the best Logos model. It demonstrates that the Weight Tying + SwiGLU recipe scales successfully to a 124M-parameter decoder-only language model and continues improving under longer training. It establishes a strong large-scale reference point for the final paper and future experiments such as context-length scaling, quantization, and distillation.